

Sets, Functions, and Cardinality

COMP2711 Discrete Mathematical Tools

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SECTION 1

Sets

Overview

Every student in a class needs a locker.

1. Which **students and lockers** are under consideration?
2. What **assignment** sends each student to a locker?
3. Can it avoid sharing and use every locker?

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sets $\longrightarrow$ **functions** $\longrightarrow$ **cardinality**

Sets

A **set** is an unordered collection of distinct objects.

$x \in A$ means “ x is an element of A .”

Throughout this course,

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ natural numbers (including 0)

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ integers

$\mathbb{R}$ real numbers: all points on the number line

Order and repetition do not create new elements.

Describing sets

Let O be the odd natural numbers below 10.

List all elements: $O = \{1, 3, 5, 7, 9\}$

Set builder $O = \{n \in \mathbb{N} \mid n < 10 \text{ and } n \text{ is odd}\}$

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A set builder needs a domain. It reads:

“the elements n in $\mathbb{N}$ for which the condition is true.”

Examples of sets

Sets can contain many kinds of objects.

Strings	$S = \{ \text{"Ada"}, \text{"Ben"}, \text{"Chen"} \}$
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Mixed objects	$M = \{7, \text{"hello"}, \{1, 2\}\}$
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In M , the number 7, the string "hello", and the set $\{1, 2\}$ are three separate elements.

Set equality

MUST UNDERSTAND A standard set is unordered and deduplicated.

Unordered

Order does not matter:

$$\{1, 3, 5\} = \{5, 3, 1\}.$$

Deduplicated

Repetition is ignored:

$$\{1, 3, 3, 5\} = \{1, 3, 5\}.$$

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$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

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Equivalently,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Membership and subsets

MUST UNDERSTAND $\in$ compares one object with a set; $\subseteq$ compares two sets.

Membership

$$x \in A$$

The object x is an element of A .

Subset

$$B \subseteq A \iff \forall x (x \in B \rightarrow x \in A)$$

Every element of B is also an element of A .

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Proper subset

$$B \subsetneq A \iff B \subseteq A \text{ and } B \neq A.$$

Thus, A contains at least one element not in B .

Membership and subset examples

Let $A = \{1, \{2, 3\}\}$.

Statement	True?	Reason
$1 \in A$		
$\{1\} \subseteq A$		
$\{2, 3\} \in A$		
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$\{2, 3\} \in A$	yes	that set is one element of A
$2 \in A$	no	being inside an element is not membership

The braces matter: 1 is a number, while $\{1\}$ is a set.

Set size

MUST UNDERSTAND For a finite set S , its size is the number of distinct elements.

The size of S , also called its **cardinality**, is written $|S|$.

Examples

Set	Size
$S = \{a, b, c\}$	$ S = 3$
$T = \{1, 1, 2\} = \{1, 2\}$	$ T = 2$

Repeating an element does not increase the size of a set.

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The empty set

$$\emptyset = \{ \}$$

It contains no elements, so

$$|\emptyset| = 0.$$

It is the **only set** of size 0.

Repeating an element does not increase the size of a set.

Power sets

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}$$

For $S = \{a, b\}$, every subset appears in the complete list:

Choice bits	00	01	10	11
Subset	$\emptyset$	$\{b\}$	$\{a\}$	$\{a, b\}$

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Two independent yes/no choices give $|\mathcal{P}(S)| = 2^2 = 4$. In general, $|S| = n$ gives $|\mathcal{P}(S)| = 2^n$.

Cartesian products

The **Cartesian product** is

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

For $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

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Each element is an **ordered pair**: the first and second positions have different roles.

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

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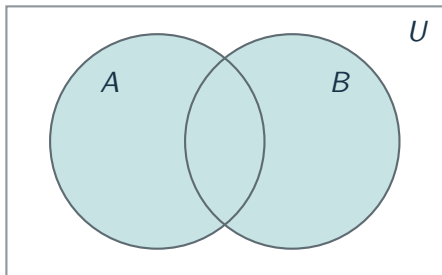
$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

If A and B are finite, $|A \times B| = |A||B|$.

SECTION 2

Set operations

Union

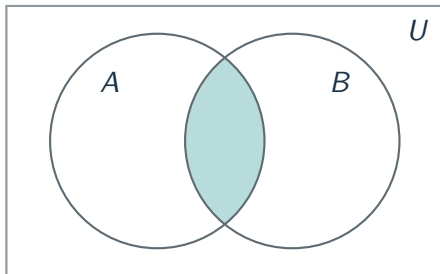


$$A \cup B$$

Shade every point that belongs to **at least one** of the sets.

$$x \in A \cup B \iff x \in A \vee x \in B.$$

Intersection



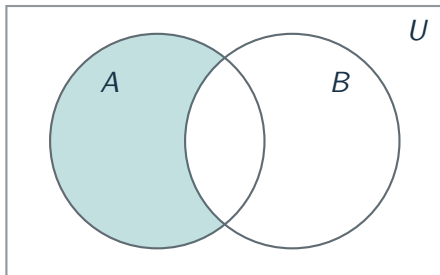
$$A \cap B$$

Shade only the points that belong to **both** sets.

$$x \in A \cap B \iff x \in A \wedge x \in B.$$

A and B are **disjoint** when $A \cap B = \emptyset$.

Difference



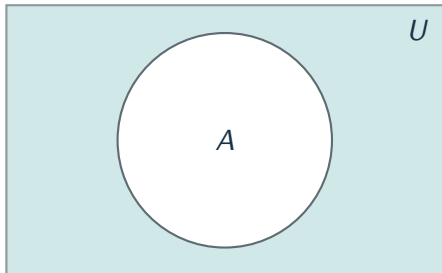
$$A \setminus B$$

Start with A , then remove every point that also belongs to B .

$$x \in A \setminus B \iff x \in A \wedge x \notin B.$$

In general, $A \setminus B \neq B \setminus A$.

Complement



$\bar{A}$

Shade everything in the universal set U that lies **outside** A .

$$x \in \bar{A} \iff x \in U \wedge x \notin A.$$

The complement depends on the chosen universe U .

Set identities and logic

Membership translates set operations into connectives:

$$\cap \longleftrightarrow \wedge, \quad \cup \longleftrightarrow \vee, \quad x \in \bar{A} \longleftrightarrow \neg(x \in A).$$

So De Morgan's logical law gives

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q \quad \implies \quad \overline{A \cup B} = \bar{A} \cap \bar{B}.$$

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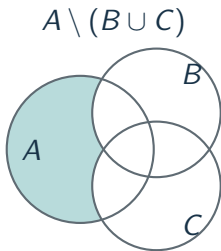
When an identity is uncertain, follow an arbitrary element.

Worked example: Venn diagram

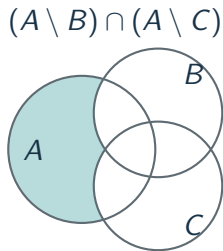
A **Venn diagram** draws sets as regions.

Compare the two sides of

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$



Remove everything in B or C from A .



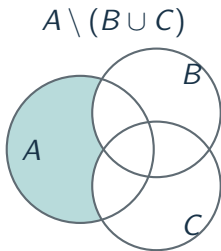
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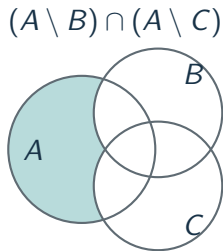
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Remove everything in B or C from A .

Both sides shade the points in A that are in neither B nor C .



Keep points in both differences.

Worked example: element proof

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

$$x \in A \setminus (B \cup C)$$

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Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

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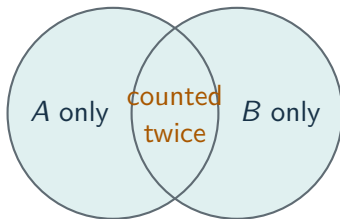
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The membership conditions agree for every x , so the sets are equal.

Inclusion–exclusion

Adding $|A| + |B|$ counts the intersection twice.



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Families of sets

For sets $\{A_i\}_{i \in I}$,

$$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \in A_i)\},$$

$$\bigcap_{i \in I} A_i = \{x \mid \forall i \in I (x \in A_i)\}.$$

Union asks for **at least one**; intersection asks for **every**.

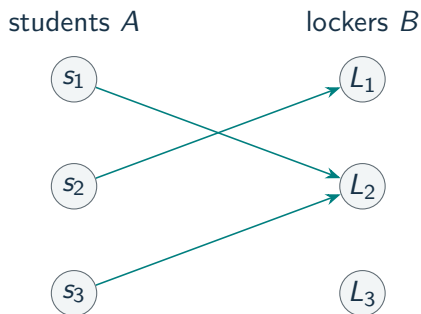
SECTION 3

Functions

Functions

MUST UNDERSTAND Every input has exactly one output.

A **function** $f : A \rightarrow B$ assigns exactly one element of B to each element of A .



Two students may share a locker. That is still a function.

Codomain and range

For a function $f : A \rightarrow B$:

Domain	A
Codomain	B
Range	elements in B <i>touched</i> by f : $f[A]$

$$f[A] = \{f(a) \mid a \in A\} \subseteq B.$$

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Example. $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$, where $f(1) = a$, $f(2) = a$, and $f(3) = c$.

Range = $\{a, c\}$. b is in the codomain, but not the range.

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Changing only the codomain can change whether a function is onto.

Recognizing functions

Each rule starts with students $\{s_1, s_2, s_3\}$.

Rule	Assignment	Function?
A	$s_1 \mapsto L_1, s_2 \mapsto L_1, s_3 \mapsto L_2$	
B	$s_1 \mapsto L_1$ and $s_1 \mapsto L_2; s_2 \mapsto L_1, s_3 \mapsto L_2$	
C	$s_1 \mapsto L_1, s_2 \mapsto L_2; s_3$ unassigned	

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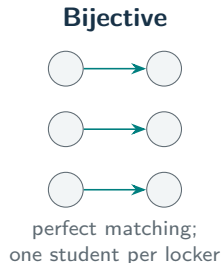
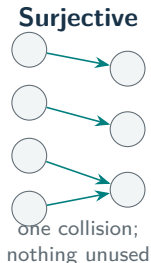
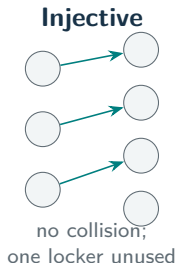
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A function may reuse outputs. It may not split or omit an input.

Injective, surjective, and bijective functions

MUST UNDERSTAND Separate collisions from unused targets.



Definitions using quantifiers

Injective:

$$\forall a_1, a_2 \in A, \quad f(a_1) = f(a_2) \rightarrow a_1 = a_2,$$

$$\text{Contrapositive: } \forall a_1, a_2 \in A, \quad a_1 \neq a_2 \rightarrow f(a_1) \neq f(a_2).$$

Equal outputs force equal inputs; distinct inputs have distinct outputs.

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Every target has at least one preimage.

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Bijjective:

$$f \text{ is bijective} \iff f \text{ is injective and surjective.}$$

Every target is reached by exactly one input.

Example: squaring on the integers

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It is **not surjective**: -1 has no integer preimage.

Its range is $\{0, 1, 4, 9, \dots\}$, not the full codomain $\mathbb{Z}$.

Changing the domain and codomain

Restrict both sides:

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If $g(m) = g(n)$, then $m^2 = n^2$; since $m, n \geq 0$, we get $m = n$.

Every target is $k^2 = g(k)$ for some $k \in \mathbb{N}$.

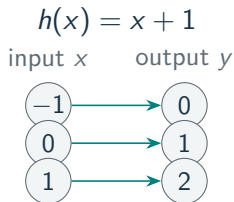
So g is **bijective**. The formula stayed the same; the sets changed.

Inverse functions

A function $f : A \rightarrow B$ has an inverse $f^{-1} : B \rightarrow A$ exactly when f is bijective.

It reverses each pair: $f^{-1}(b) = a \iff f(a) = b$.

Example. Let $h : \mathbb{Z} \rightarrow \mathbb{Z}$ be $h(x) = x + 1$.



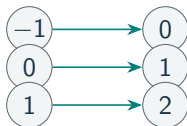
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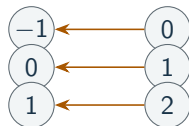
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$h(x) = x + 1$
input x output y



$h^{-1}(y) = y - 1$
output x input y



Composition

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$$f \circ g : A \rightarrow C, \quad (f \circ g)(a) = f(g(a)).$$

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Let $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$.

$$(f \circ g)(x) = 2(x + 1) = 2x + 2.$$

Apply g first, then apply f .

Composition laws

Associative. Whenever the compositions are defined,

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The grouping changes, but the order remains h , then g , then f .

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Generally not commutative. Let $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$.

$$(f \circ g)(x) = 2x + 2, \quad (g \circ f)(x) = 2x + 1.$$

Therefore $f \circ g \neq g \circ f$. Changing the order can change the function.

Floor and ceiling

$\lfloor x \rfloor$ = greatest integer at most x , $\lceil x \rceil$ = least integer at least x .



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$\lfloor x \rfloor$ = greatest integer at most x , $\lceil x \rceil$ = least integer at least x .



$$-3 \leq -2.7 < -2, \quad \lfloor -2.7 \rfloor = -3, \quad \lceil -2.7 \rceil = -2.$$

Counting functions

Let $|A| = n$ and $|B| = m$.

Each input has m independent choices, so

$$\#\{f : A \rightarrow B\} = m^n.$$

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For an injection, outputs cannot be reused:

$$\#\{\text{injections } A \rightarrow B\} = \begin{cases} m(m-1) \cdots (m-n+1), & m \geq n, \\ 0, & m < n. \end{cases}$$

Between two n -element sets: n^n functions and $n!$ bijections.

Set practice

Let

$$A = \{1, 2\}, \quad B = \{2, 3\}.$$

Compute:

1. $A \cup B$
2. $A \cap B$
3. $A \setminus B$
4. $\mathcal{P}(A)$
5. $A \times B$

Write every resulting set by listing all of its elements.

Set practice solution

For $A = \{1, 2\}$ and $B = \{2, 3\}$:

$$A \cup B = \{1, 2, 3\},$$

$$A \cap B = \{2\},$$

$$A \setminus B = \{1\},$$

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\},$$

$$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}.$$

Function practice

Let $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ be

$$f(n) = n + 3, \quad g(n) = 2n.$$

1. Find $(f \circ g)(n)$ and $(g \circ f)(n)$.
2. Decide whether each function is injective, surjective, or bijective.
3. If a function is bijective, find its inverse.

Remember that both the domain and codomain are $\mathbb{Z}$.

Function practice solution

$$(f \circ g)(n) = f(2n) = 2n + 3,$$
$$(g \circ f)(n) = g(n + 3) = 2n + 6.$$

Function	Classification and reason
f	Bijjective: adding 3 gives a distinct output for every input and reaches every integer.
g	Injective but not surjective: its outputs are even, so it misses every odd integer.

Since f is bijective,

$$f^{-1}(n) = n - 3.$$

SECTION 4

Cardinality

Comparing cardinalities

For a finite set, cardinality is found by counting its elements. For an infinite set, ordinary counting never finishes, so we compare sets by matching their elements.

Comparison	Function evidence
$ A = B $	a bijection $A \rightarrow B$
$ A \leq B $	an injection $A \rightarrow B$
$ A < B $	an injection exists, but no bijection exists

These definitions work for finite and infinite sets.

Injection and strict inequality

Consider the shift

$$s : \mathbb{N} \rightarrow \mathbb{N}, \quad s(n) = n + 1.$$

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It is injective and not surjective: 0 is missed.

But $|\mathbb{N}| = |\mathbb{N}|$; the identity map is a bijection.

Wrong: “injective but not bijective” $\Rightarrow |A| < |B|$.

Correct: strict inequality also requires that no bijection exists.

Hilbert's hotel

Rooms $0, 1, 2, \dots$ are all occupied.

Hilbert's hotel

Rooms $0, 1, 2, \dots$ are all occupied.

For one new guest:

$$\text{old guest in room } n \mapsto n + 1.$$

Room 0 becomes free.

For countably many new guests:

$$\text{old } n \mapsto 2n, \quad \text{new } k \mapsto 2k + 1.$$

Every guest gets exactly one room, and no rooms are shared.

Countable sets

MUST UNDERSTAND Countable means finite or listable by $\mathbb{N}$.

A set is **countable** when it is finite or countably infinite.

An infinite set S is countably infinite when there is a bijection

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Equivalently, a list has **no omissions and no repetitions**:

$$e(0), e(1), e(2), \dots$$

Its cardinality is written $|S| = \aleph_0$.

The integers are countable

List them by alternating signs:

$$0, 1, -1, 2, -2, 3, -3, \dots$$

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A precise enumeration $e : \mathbb{N} \rightarrow \mathbb{Z}$ is

$$e(0) = 0, \quad e(2k - 1) = k, \quad e(2k) = -k \quad (k \geq 1).$$

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Positive k occurs at index $2k - 1$; negative $-k$ occurs at $2k$.

Every integer occurs once, so $\mathbb{Z}$ is countably infinite.

The rational numbers are countable

Every positive rational is p/q for positive integers p, q .

$p + q$	Pairs (p, q)	New values p/q
2	(1, 1)	1
3	(1, 2), (2, 1)	$1/2, 2$
4	(1, 3), (2, 2), (3, 1)	$1/3, 1$ (repeat), 3
5	(1, 4), (2, 3), (3, 2), (4, 1)	$1/4, 2/3, 3/2, 4$

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Continue by constant $p + q$ and skip any value already listed. Every positive rational eventually appears.

If $q_1, q_2, \dots$ is this positive list, then

$$0, q_1, -q_1, q_2, -q_2, \dots$$

lists every rational exactly once.

Products of countable sets

Suppose $A = \{a_0, a_1, \dots\}$ and $B = \{b_0, b_1, \dots\}$ are countably infinite.

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In particular, $\mathbb{Q} \times \mathbb{Q}$ is countable.

Finite factors are immediate.

Uncountable sets

An infinite set is **uncountable** when no sequence $s_1, s_2, s_3, \dots$ contains every element.

Q: a complete list exists

position	rational
1	$\longrightarrow q_1$
2	$\longrightarrow q_2$
3	$\longrightarrow q_3$
$\vdots$	$\vdots$

$\mathbb{R}$: every proposed list fails

position	listed real
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$$|\mathbb{Q}| = \aleph_0 < |\mathbb{R}| = \mathfrak{c}.$$

Next: construct r^* from the proposed list itself.

Cantor's diagonal argument

MUST UNDERSTAND Build one object that differs from item i at position i .

Suppose $r_i = 0.d_{i1}d_{i2}d_{i3} \dots$ lists every number in $(0, 1)$. Choose expansions that do not end in infinitely many 9s.

For a concrete proposed list, focus on the diagonal:

$$r_1 = 0.2379 \dots$$

$$r_2 = 0.4459 \dots$$

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Choose

$$c_i = \begin{cases} 4, & d_{ii} \neq 4, \\ 5, & d_{ii} = 4, \end{cases} \quad r = 0.c_1c_2c_3c_4 \dots$$

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The constructed r differs from r_i at decimal position i .

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This is **Cantor diagonalization**.

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$$|A| \leq |B| \text{ and } |B| \leq |A| \implies |A| = |B|.$$

Compare $(0, 1)$ and $(0, 1]$:

$$\begin{aligned} (0, 1) &\hookrightarrow (0, 1], & x &\mapsto x, \\ (0, 1] &\hookrightarrow (0, 1), & x &\mapsto x/2. \end{aligned}$$

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Hence $|(0, 1)| = |(0, 1]|$.

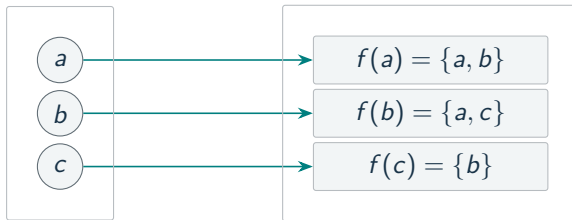
Cantor's theorem

$$|S| < |\mathcal{P}(S)| \quad \text{for every set } S.$$

That is, the power set of S is “bigger” than S in cardinality.

$$S = \{a, b, c\}$$

outputs in $\mathcal{P}(S)$



This is one possible f : each element of S maps to **one subset of S** .

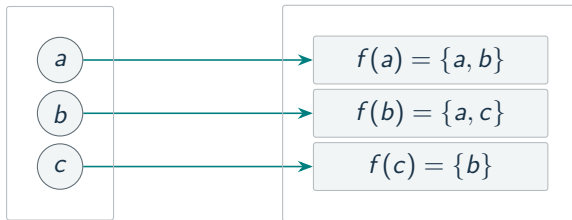
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The injection $x \mapsto \{x\}$ gives $|S| \leq |\mathcal{P}(S)|$. For strict inequality, show that **no $f : S \rightarrow \mathcal{P}(S)$ reaches every subset; equivalently, no surjective f exists.**

Constructing the diagonal subset

For any $f : S \rightarrow \mathcal{P}(S)$, define

$$D = \{x \in S \mid x \notin f(x)\} \in \mathcal{P}(S).$$

It collects the elements that do not belong to their own assigned subsets.

This works for every f . No surjectivity assumption yet.

Input	Assigned subset	Self-membership	Decision for D
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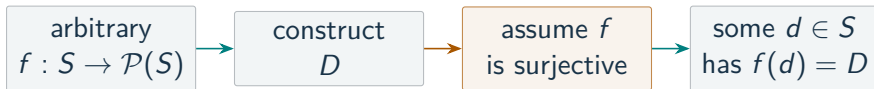
For this example, $D = \{b, c\}$; it is not one of the three outputs.

The diagonal subset cannot be reached



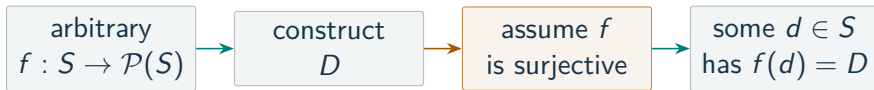
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$$\boxed{d \in D} \iff d \notin f(d) \iff \boxed{d \notin D}.$$

Either answer to “is d in D ?” contradicts itself.

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Either answer to “is d in D ?” contradicts itself.

Therefore D is not reached by f . No function $S \rightarrow \mathcal{P}(S)$ is surjective, so $|S| < |\mathcal{P}(S)|$.

The continuum hypothesis

The size of the real numbers is the **continuum**:

$$\aleph_0 < \mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}.$$

subsets of $\mathbb{N}$ $\longleftrightarrow$ infinite bit strings $\longleftrightarrow$ reals

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The **continuum hypothesis** asks:

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In words: is there no cardinality strictly between the sizes of $\mathbb{N}$ and $\mathbb{R}$?

Optional: CH is independent of ZFC

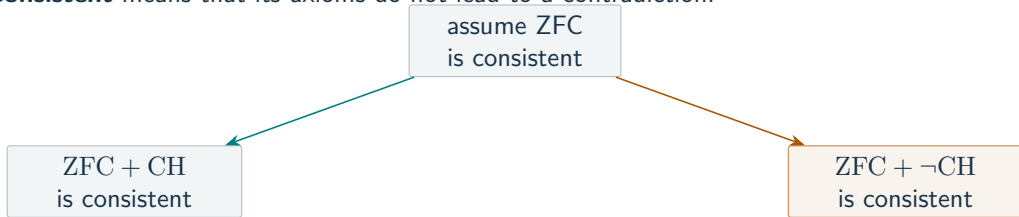
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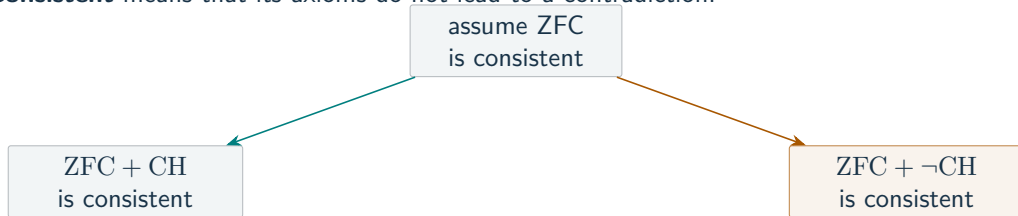


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Gödel (1940) ZFC cannot disprove CH, assuming ZFC is consistent.

Cohen (1963) ZFC cannot prove CH, assuming ZFC is consistent.

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Therefore some functions $\mathbb{N} \rightarrow \{0, 1\}$ are uncomputable. This proves existence; it does not identify the halting problem.

SECTION 5

Worked problems

Worked problem: classifying a function

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Decide whether f is injective and surjective. If an inverse exists, find it.

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Plan:

1. For injection, start from equal outputs.
2. For surjection, start from an arbitrary target $y \in \mathbb{Z}$.

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$$y = 3n - 2 \iff n = \frac{y + 2}{3}.$$

For $y = 0$, this is not an integer. So f is not surjective.

No inverse $\mathbb{Z} \rightarrow \mathbb{Z}$ exists. Restrict the codomain to $\{3n - 2 \mid n \in \mathbb{Z}\}$ to obtain a bijection.

Worked problem: infinite subsets

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$E = \{0, 2, 4, \dots\}$ is a proper subset of $\mathbb{N}$. Therefore $|E| < |\mathbb{N}|$.

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For infinite sets, proper containment alone does not prove smaller cardinality.

Summary

1. Sets are determined by membership; set algebra is logic applied to membership.
2. Functions transform elements. Injection, surjection, and bijection answer different matching questions.
3. Cardinality compares sizes by functions. Diagonalization proves $\aleph_0 < \mathfrak{c}$; CH asks whether $\mathfrak{c} = \aleph_1$.

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Next: **mathematical induction** proves a claim for every natural number by linking one case to the next.